Beyond Weak Supervision: MLLMs-Guided Graded Knowledge Distillation for Unsupervised Camouflaged Object Detection

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Abstract

Most Camouflaged Object Detection (COD) methods rely on costly pixel-level annotations. Recent studies have adopted unsupervised COD (UCOD) to eliminate labeling costs, but still suffer from two issues: 1) insufficient supervision, leading to reliance on self-supervised backbone DINO and reduced model flexibility; and 2) ineffective use of pseudo-labels, which widens the performance gap with supervised methods and limits real-world applicability. In this paper, we propose a novel teacher-student framework for UCOD to address these two issues. To tackle the lack of supervision, we build a powerful teacher model by integrating Multimodal Large Language Models (MLLMs) and the Segment Anything Model (SAM) to generate high-quality pseudo-labels. However, the teacher model faces two challenges: 1) suboptimal performance of MLLMs in COD, and 2) cascading errors. To address these challenges, we first propose a Camouflaged-Aware Chain-of-Thought (CA-CoT) for MLLMs. CA-CoT guides MLLMs through step-by-step reasoning to simulate human perceptual processes, thereby enhancing their performance in COD. Subsequently, we design a Graded Mask Evaluator (GME) to mitigate cascading errors, which evaluates and grades the quality of masks generated by SAM, and then filters out the low-quality masks to provide more reliable supervision. To better leverage pseudo-labels, we propose Graded Knowledge Distillation (GKD), which adaptively enhances distillation at both image and pixel levels based on pseudo-label quality. Extensive experiments show that our method outperforms existing UCOD approaches by a large margin. Notably, our method also achieves good performance under zero-shot settings. Code will be released.

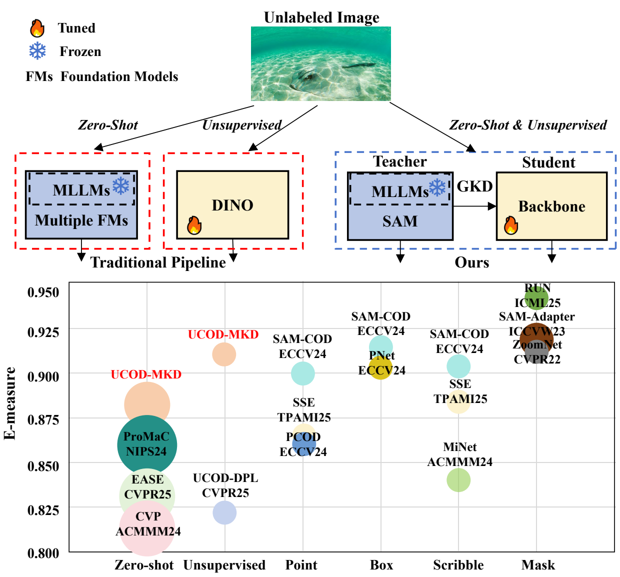

Figure 1. **Top:** Pipeline comparison between traditional unsupervised, traditional zero-shot, and our COD methods. **Bottom:** Performance comparison of COD methods under varying supervision. Larger circles indicate models with more parameters. Our method outperforms UCOD and zero-shot methods, and even delivers competitive results compared to WSCOD methods.

1. Introduction

Camouflaged Object Detection (COD) aims to identify objects that are concealed within their surroundings [11, 20, 34]. In recent years, this task has gained growing attention in the computer vision community and facilitates valuable real-life applications, such as search and rescue [14], species discovery [35], and medical image analysis [12, 13].

Although fully supervised methods have achieved remarkable progress in COD tasks [17, 20, 28, 30, 49], they heavily rely on pixel-level annotations, which makes the labeling process extremely costly. To reduce annotation costs, some studies [5, 6, 19, 21] have adopted weak supervision instead of full supervision. While this approach alleviates the labeling burden to some extent, it still requires per-image manual labeling and thus fails to eliminate annotation costs entirely. To further eliminate annotation costs, recent studies [37, 45, 48] have begun exploring unsupervised

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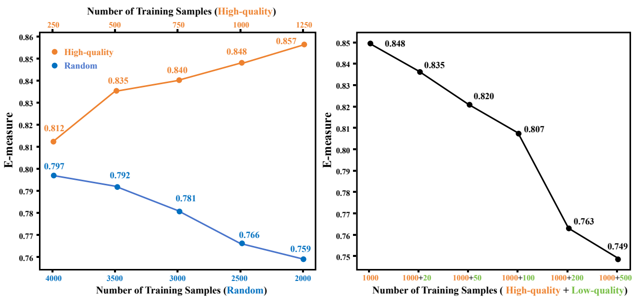

Figure 2. COD distillation follows the principle of quality over quantity. **Left:** The model's performance improves with fewer high-quality samples compared to large random samples. **Right:** The model's performance degrades rapidly with the addition of low-quality samples, even in small amounts (∼2%), and once their proportion exceeds ∼15%, it severely disrupts learning and causes a sharp performance drop.

camouflaged object detection (UCOD) methods.

However, as shown in Fig. 1, existing UCOD methods still suffer from two key limitations: 1) An over-reliance on the powerful self-supervised backbone DINO [4], which reduces the flexibility of the models; 2) A significant performance gap compared to supervised models, which restricts their applicability in real-world scenarios. To better understand these limitations, we analyzed existing UCOD methods and identified two causes: 1) *A lack of ability to extract effective supervision signals* from unlabeled data, forcing methods to rely heavily on DINO while still failing to bridge the gap with supervised approaches. 2) *Limited capability in leveraging pseudo-labels efficiently*, which further amplifies the performance shortfall by preventing models from fully exploiting the available unlabeled data.

In this paper, we propose a teacher-student model called UCOD-MKD. To address the lack of effective supervision signals, UCOD-MKD constructs a powerful teacher model by integrating Multimodal Large Language Models (MLLMs) and the Segment Anything Model (SAM) [27] to generate high-quality supervision signals. However, enabling the teacher model to generate high-quality supervision signals is no trivial task, as it faces two main challenges: 1) *Poor visual grounding performance of MLLMs*: Owing to the lack of training on camouflaged data, MLLMs are prone to hallucination and instability in the challenging COD task, which degrades localization quality. 2) *Cascading error issue*: As shown in Fig. 7, integrating multiple foundation models often leads to error accumulation, which results in extremely low-quality masks. As shown in Fig. 2, COD distillation follows a quality-over-quantity principle. Thus, such extremely low-quality masks can significantly degrade distillation performance, placing stringent requirements on the design of the teacher model to ensure the reliability of mask quality. To mitigate the aforementioned issues, we first propose Camouflaged-Aware Chain-of-Thought (CA-CoT), an extension of standard prompt engineering that incorporates step-by-step reasoning to simulate human perceptual processes. CA-CoT effectively mitigates hallucinations and localization jitter, significantly improving visual grounding accuracy in COD scenarios.

Subsequently, we devise a Graded Mask Evaluator (GME) to mitigate the cascading error issue. GME evaluates the quality of SAM-generated masks based on three tailored metrics: similarity, fragmentation, and the number of edge truncations. Then, GME performs quality stratification and filters out low-quality masks to help safeguard the distillation process. With the integration of CA-CoT and GME, UCOD-MKD enables the teacher model to produce high-quality supervision, effectively eliminating reliance on the DINO backbone and enhancing overall performance.

The pseudo-labels provided by the teacher model are used to train the student model through a distillation process. Existing methods typically adopt an **equal treatment** strategy for all samples or pixels during distillation. However, since pseudo-labels cannot achieve the same level of accuracy as ground-truth labels, there are often significant variations in precision between different samples and even among pixels within the same sample. Such indiscriminate treatment leads to underutilization of pseudo-labels and adversely affects the effectiveness of distillation. To address the limited capability of existing distillation methods in leveraging pseudo-labels, we propose a Graded Knowledge Distillation (GKD) strategy. By extracting prior knowledge from the teacher model, GKD performs differentiated distillation on pseudo-labels at both the image level and the pixel level, thereby fully unlocking the potential of pseudo-labels and enhancing the overall distillation performance.

Our contributions are summarized as follows:

- We propose a novel UCOD model called UCOD-MKD, based on a teacher-student framework. To the best of our knowledge, this is the first model in the COD domain capable of simultaneously supporting both zero-shot and unsupervised training settings.
- We construct a powerful teacher model by integrating MLLMs and SAM to generate high-quality supervision. The teacher model is designed with a CA-CoT to guide MLLMs' localization reasoning via simulated human perception, and a GME for SAM to grade output masks with defined metrics and filter out low-quality ones.
- We propose GKD, which extracts prior knowledge from the teacher model and conducts differentiated distillation at image and pixel levels, thereby fully exploiting pseudo-labels to improve distillation performance.
- Experimental results show that our method significantly outperforms existing UCOD approaches and achieves performance competitive with weakly-supervised COD methods. It also performs well in the zero-shot setting.

2. Related Work

Camouflaged Object Detection. Camouflaged Object Detection (COD) focuses on detecting camouflaged objects within an image. Most existing studies primarily focus on fully supervised COD methods [20, 26, 30, 34, 49]. However, these methods heavily depend on pixel-level annotations, making the labeling process extremely costly. To reduce annotation costs, some studies [5, 6, 19, 21, 32] employ weakly supervised labels for COD, but still require manual labeling per image. Beyond weak supervision, recent work [8, 40, 45, 48] investigates unsupervised COD (UCOD), which requires no manual annotations whatsoever. UCOS-DA [48] introduces the first UCOD framework by adopting self-supervised ViT-based domain adaptation. UCOD-DPL [45] dynamically fuses pseudo-labels via APM while addressing ambiguity via dual-branch adversarial learning and Look-Twice refinement. CVP [40] and EASE [8] achieve zero-shot COD by incorporating multiple foundation models. CVP [40] proposes a chain of visual perception (CoVP) to improve the perceptual capabilities of MLLMs in COD. EASE [8] introduces environment-feature inversion for COD through its diffPro library and three retrieval mechanisms. The aforementioned unsupervised training COD methods have two issues: 1) *An insufficient ability to obtain effective supervision signals*: This leads to a heavy reliance on self-supervised backbone DINO, limiting the model's flexibility and effectiveness. 2) *A limited capability in utilizing pseudo-labels*: This exacerbates the performance gap compared to supervised models and restricts practical use.

Multimodal Large Language Models for COD. Multimodal Large Language Models (MLLMs) achieve unified vision-language understanding through joint visual-textual token alignment. These MLLMs, like GPT-4V [1], LLaVA-1.5 [29] and Qwen2.5-VL [2], demonstrate unparalleled adaptability across diverse tasks. Recent studies have explored using MLLMs for COD tasks. EASE [8] utilizes MLLMs to extract category-level textual information from images. GenSAM [23] proposes Cross-modal Chains of Thought Prompting to map text prompt onto image heatmaps using MLLMs, acquiring visual prompts. ProMaC [24] proposes Multi-scale Chain of Thought Prompting to generate instance-specific prompts via hallucination refinement, but it relies on multiple image inputs to construct prompts, which significantly increases the computational overhead of MLLMs. CVP [40] proposes CoVP to improve MLLMs' perception in COD, but it mainly reinforces camouflage-related concepts in prompts without step by step reasoning, lacking a true chain-of-thought process. Unlike prior zero-shot methods, our approach is the first to leverage MLLMs for unsupervised COD training, introducing a Camouflaged-Aware Chain-of-Thought driven purely by textual prompts to enhance reasoning, with almost no impact on computational cost or inference speed.

Segment Anything Model for COD. Segment Anything Model (SAM) [27] excels in traditional segmentation tasks [15, 16]. However, recent studies [7, 39] reveal its limitations in COD tasks, where it tends to generate low-quality masks with prompt-based inputs (e.g., points or boxes). To mitigate this issue, recent weakly supervised COD (WSCOD) methods employ distinct refinement strategies: SEE [19] refines masks through a multi-augmentation ensemble strategy, and SAM-COD [6] selects the best mask using a response filter and semantic matcher. In contrast to WSCOD, zero-shot approaches rely on auxiliary models (e.g., MLLMs, CLIP) for initial prompt generation. Given that these prompts may contain errors, it is critical to account for the potential impact of such erroneous prompts. CVP [40] introduces DINOv2 [33] to enhance the point prompts from MLLMs. ProMaC [23] introduces CLIP [36] to select SAM-generated masks. But the drawbacks of those are also obvious: 1) *Additional computational overhead*: SEE require multiple inference passes, SAM-COD needs extra training, while CVP and ProMaC depend on additional foundation models. 2) *Lack of hierarchical mask quality*: These methods optimize single-sample masks but ignore relative quality differences across samples.

Knowledge Distillation. Knowledge Distillation (KD) [3, 4, 22] has been primarily designed to train a small network to mimic the output of a larger network to compress models [41, 42]. Existing UCOD methods [37, 45, 48] typically perform traditional distillation by treating all samples or pixels equally. In contrast, SAM-COD [6] argues that traditional distillation methods are ill-suited for WSCOD and proposes a prompt-adaptive knowledge distillation to mitigate the effects of inaccurate masks. Different from the aforementioned method: 1) We identify a quality over quantity principle, where extremely low-quality pseudo-labels are filtered out, enabling selective distillation instead of full-scope distillation. 2) We extract prior knowledge from the teacher model and conduct differentiated distillation at image and pixel levels.

3. Methodology

3.1. Overview

As shown in the top-left of Fig. 3, we adopt a teacher-student framework to achieve unsupervised camouflaged object detection (UCOD), which consists of a teacher model and a student model. The teacher model comprises a Multimodal Large Language Model (MLLM) and a Segment Anything Model (SAM), both with fixed parameters. The student model consists of a trainable backbone network. Given an input image, the teacher model employs Camouflage-Aware Chain-of-Thought (CA-CoT) prompting to guide the MLLM through step-wise reasoning, gen-

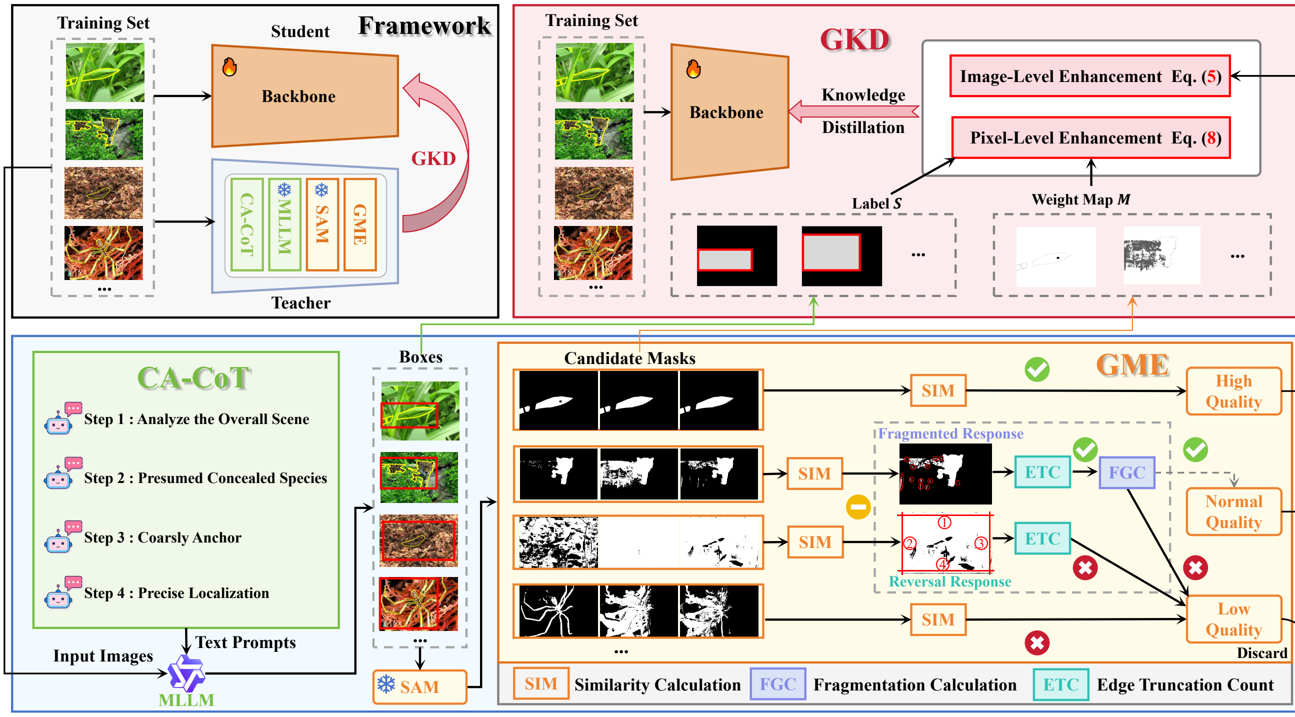

Figure 3. Overview of our proposed UCOD-MKD method. The top-left figure illustrates the framework, which follows a teacher-student architecture. The teacher model consists of an MLLM and SAM, and generates high-quality graded masks through CA-CoT and GME. These masks are then used to distill knowledge into the student model via GKD.

erating localization bounding boxes. The bounding boxes provide prompts to SAM for generating candidate masks, whose quality is then evaluated and graded by the Graded Mask Evaluator (GME). The student model then adaptively distills these graded masks through Graded Knowledge Distillation (GKD).

3.2. Camouflage-Aware Chain-of-Thought

We use MLLM to obtain initial localization information for camouflaged objects. However, visual localization of MLLM is prone to hallucinations and jitter, which becomes more pronounced in challenging camouflage scenarios and severely affects localization quality. To enhance MLLM's localization performance in camouflage scenarios, we propose Camouflage-Aware Chain-of-Thought (CA-CoT), a specialized Chain-of-Thought (CoT) framework for COD that incorporates human perceptual mechanisms. As shown in Fig. 4, the CA-CoT guidance process consists of two stages: 1) **Progressive reasoning from global scene to local objects** (**STEP 1 & 2**). By mimicking human reasoning logic, MLLM is guided to first scan the input image to establish a holistic understanding of the overall scene (e.g., forest, desert) (STEP 1), and then infer potential types of camouflaged objects (e.g., snakes, lizards) based on scene characteristics (STEP 2). 2) **Progressive perception from coarse to fine localization** (**STEP 3, 4 & 5**). Leveraging clues from stage 1, MLLM is further guided to mimic human perception. Initially, MLLM coarsely anchors potential objects by utilizing color and texture similarities between the camouflaged object and background (STEP 3). Then, mimicking human detailed observation, MLLM is guided to focus on the geometric features (e.g., boundaries, shapes) of the camouflaged object to define its full extent, ultimately achieving precise localization (STEP 4 & 5).

3.3. Graded Mask Evaluator

After obtaining initial bounding boxes through MLLM, we use SAM to convert these boxes into pixel-level pseudo-labels. Although CA-CoT significantly improves the localization accuracy of MLLMs, there are inevitably some inaccurate bounding boxes, as shown in Fig. 7. Due to the cascading error, these inaccurate boxes can lead to extremely low-quality masks generated by SAM. More importantly, the distillation process in COD follows the principle of *quality over quantity*, meaning that these extremely low-quality masks greatly reduce the effectiveness of knowledge distillation. To avoid this issue, we design a Graded Mask Evaluator (GME). As shown in Fig. 3, GME assesses the quality of SAM-generated masks, assigns each a quality grade, and then removes those graded as low-quality.

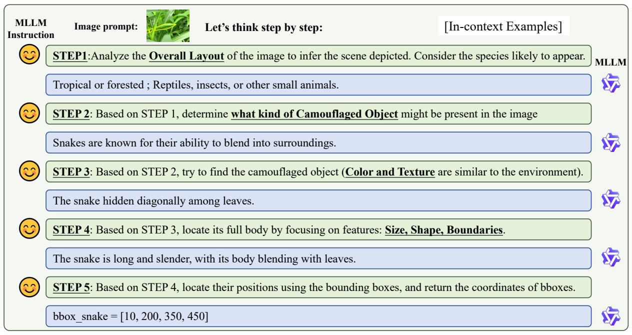

Figure 4. Details of the proposed CA-CoT.

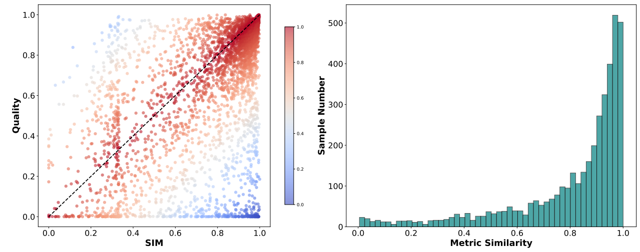

Figure 5. There is a strong correlation between the SIM of candidate masks and their quality. **Left:**Visualization of SIM and corresponding quality for all candidate masks. Points closer to the diagonal indicate a higher alignment between SIM and quality. **Right:** Distribution of samples across different levels of alignment between SIM and quality.

Specifically, SAM uses the box as a prompt to generate candidate masks:

$$\{V_j^k | k = 1, 2, 3\} = SAM(I_j, Box_j), \tag{1}$$

where V_j^k denotes the k-th candidate mask of the j-th sample in the training set. We found that inaccurate bounding boxes lead to unstable segmentation by SAM (i.e., low similarity among its candidate masks), and vice versa. This is because SAM struggles to determine whether to segment the background when given an inaccurate box prompt. Consequently, there exists a strong correlation between the similarity among candidate masks and their segmentation quality, as illustrated in Fig. 5. Motivated by this, we evaluate segmentation quality based on the similarity among candidate masks, computed with IoU and SSIM [44], two widely used similarity metrics, as follows:

$$\text{SIM}(V_j^{k_1}, V_j^{k_2}) = \frac{1}{2}(\text{IoU}(V_j^{k_1}, V_j^{k_2}) + \text{SSIM}(V_j^{k_1}, V_j^{k_2})), \tag{2}$$

where $V_j^{k_1}$ and $V_j^{k_2}$ denote two distinct candidate masks of the j-th sample. Subsequently, the final similarity metric is obtained by averaging the pairwise similarities between candidate masks:

$$S_j = \frac{1}{3} \sum_{k_1 < k_2} \text{SIM}(V_j^{k_1}, V_j^{k_2}), \tag{3}$$

where S_j represents the similarity metric of the j-th sample. We perform mask grading by defining a pass grade (60%) and an excellent grade (90%):

$$\text{Q}_j = \begin{cases} 0, & \text{if } S_j < 0.6 \\ 1, & \text{if } 0.6 \le S_j < 0.9 \\ 2, & \text{if } S_j \ge 0.9 \end{cases} \tag{4}$$

where Q_j denotes the quality grade of the j-th sample, with 0 for low-quality, 1 for normal-quality, and 2 for high-quality. During distillation, we exclude low-quality masks and retain high-quality masks. Since the masks at the normal-quality grade vary significantly in quality, they cannot be directly discarded or retained. To refine the selection within this grade, we analyze failure cases. Through observation, we found that the poor masks within the normal-quality grade exhibit two main characteristics: fragmented responses and reversal responses, as shown in Fig. 3.

Therefore, based on the above characteristics, we further re-evaluate the normal-quality grade masks. For the reversal response, we apply **Edge Truncation Count** (ETC), which counts the number of response masks truncated by boundaries and filters out masks with an excessive number of truncations. For the fragmented responses, we perform **Fragmentation Calculation** (FGC), counting the number of connected components and filtering out masks with an excessive number of connected components. Only the masks that pass both ETC and FGC are retained in the normal-quality grade; the rest are reclassified as low-quality masks.

3.4. Graded Knowledge Distillation

The pseudo-labels provided by the teacher model are used to train the student model through a knowledge distillation (KD) process. Traditional KD treats different samples or pixels equally, which leads to underutilization of pseudo-labels. To fully exploit the potential of pseudo-labels, we propose Graded Knowledge Distillation (GKD). As shown in Fig. 3, GKD enhances the KD process by extracting prior knowledge from the teacher model and applying targeted enhancements at both the image-level and pixel-level.

Image-Level Enhancement. We extract pseudo-label quality grades from GME and use them to guide differentiated distillation strategies across samples. For low-quality grade samples, although the pseudo-labels are of little value, the image content itself remains valuable. Therefore, we apply self-knowledge distillation (SKD) [4] to these samples. For normal-quality grade samples, we adopt the commonly used cross-entropy (CE) loss. As for high-quality grade samples, we provide stronger supervision by incorporating multiple

Methods	Backbones	Sup.	CAMO				COD10K				NC4K			
			MAE↓	S_m↑	E_m↑	F_β^w↑	MAE↓	S_m↑	E_m↑	F_β^w↑	MAE↓	S_m↑	E_m↑	F_β^w↑
Fully-Supervised Methods														
SINet [11] CVPR'20	ResNet50	F	0.092	0.745	0.804	0.644	0.043	0.776	0.864	0.631	0.058	0.808	0.871	0.723
UGTR [46] ICCV'21	ResNet50	F	0.086	0.784	0.822	0.684	0.036	0.817	0.852	0.666	0.052	0.839	0.874	0.747
ZoomNet [34] CVPR'22	ResNet50	F	0.066	0.820	0.892	0.752	0.029	0.838	0.911	0.729	0.043	0.853	0.896	0.784
FEDER [18] CVPR'23	Res2Net50	F	0.066	0.836	0.897	0.806	0.029	0.844	0.911	0.736	0.042	0.862	0.913	0.823
SAM-Ada. [7] ICCVW'23	VIT-H	F	0.070	0.847	0.873	0.765	0.025	0.883	0.918	0.801	-	-	-	-
HitNet [25] AAAI'23	PVT V2	F	0.060	0.834	0.892	0.801	0.027	0.847	0.922	0.798	0.042	0.858	0.911	0.825
CamoFocus [26] WACV'24	PVT V2	F	**0.044**	0.870	0.924	0.842	0.022	0.868	0.931	0.802	0.031	0.886	0.932	0.853
RUN [20] ICML'25	PVT V2	F	0.045	**0.877**	**0.934**	**0.853**	**0.021**	**0.878**	**0.941**	**0.808**	**0.030**	**0.892**	**0.940**	**0.866**
Weakly-supervised Methods														
CRNet [21] AAAI'23	ResNet50	S	0.092	0.735	0.815	0.641	0.049	0.733	0.832	0.576	0.063	0.775	0.855	0.688
MiNet [32] MM'24	ResNet50	S	0.091	0.750	0.840	0.669	0.049	0.749	0.840	0.596	0.061	0.793	0.869	0.709
SSE [19] TPAMI'25	ResNet50	S	0.090	0.765	0.826	0.742	0.036	0.807	0.883	0.724	0.051	0.836	0.891	0.805
SAM-COD [6] ECCV'24	PVT V2	S	0.060	0.836	0.903	0.779	0.029	0.833	0.904	0.728	0.039	0.859	0.912	0.795
PCOD [5] ECCV'24	PVT V2	P	0.074	0.798	0.872	0.721	0.042	0.784	0.859	0.650	0.051	0.822	0.889	0.748
SAM-COD [6] ECCV'24	PVT V2	B	0.062	0.837	0.901	0.786	**0.028**	**0.842**	**0.914**	**0.745**	0.037	**0.867**	0.923	0.813
PNet [47] ECCV'24	PVT V2	B+F	**0.051**	**0.852**	**0.922**	**0.831**	0.031	0.828	0.903	0.739	**0.037**	0.864	**0.926**	**0.824**
Unsupervised Methods														
FOUND [38] CVPR'23	DINO V1	U	0.127	0.701	0.784	0.606	0.086	0.689	0.740	0.513	0.085	0.755	0.819	0.656
UCOS-DA [48] ICCVW'23	DINO V1	U	0.129	0.685	0.782	0.584	0.085	0.670	0.751	0.482	0.084	0.741	0.824	0.637
SdalsNet [37] AAAI'25	DINO V1	U	0.117	0.696	-	-	0.071	0.696	-	-	0.084	0.738	-	-
UCOD-DPL [45] CVPR'25	DINO V1	U	0.108	0.706	0.801	0.621	0.059	0.727	0.822	0.577	0.074	0.761	0.851	0.680
UCOD-MKD (Ours)	ResNet50	U	0.085	0.764	0.833	0.701	0.039	0.781	0.869	0.684	0.051	0.821	0.884	0.757
UCOD-MKD (Ours)	PVT V2	U	**0.071**	**0.809**	**0.875**	**0.753**	**0.031**	**0.835**	**0.908**	**0.740**	**0.040**	**0.857**	**0.918**	**0.803**
Zero-shot Methods														
GenSAM [23] AAAI'24	BLIP-$2_{6.7B}$, CLIP, SAM	Z	0.117	0.730	-	0.655	0.069	0.731	-	0.584	0.087	0.754	-	0.665
CVP [40] MM'24	Shikra$_{7B}$, DINO V2, SAM-HQ	Z	0.100	0.749	-	0.680	0.065	0.733	-	0.592	0.082	0.768	-	0.681
EASE [8] CVPR'25	LLaVA-1.5_{7B}, DINO V2, SD V1.5	Z	0.166	0.653	0.737	0.563	0.109	0.673	0.732	0.514	0.108	0.728	0.790	0.633
ProMaC [24] NIPS'24	LLaVA-1.5_{13B}, CLIP, SAM, SD V2	Z	**0.098**	**0.775**	**0.832**	**0.698**	0.045	0.795	0.861	0.624	0.073	0.779	0.845	0.707
UCOD-MKD (Ours)	Qwen2.5-VL$_{3B}$, SAM	Z	0.101	0.738	0.786	0.652	**0.036**	**0.830**	**0.882**	**0.755**	**0.059**	**0.823**	**0.866**	**0.766**

Table 1. Comparison of our methods with recent methods. "F", "S", "B", "P", "Z" and "U" denote fully-supervised, scribble-supervised, box-supervised, point-supervised, zero-shot and unsupervised methods, respectively. "B+F" indicates hybrid supervision (99% B + 1% F). **Bold** indicates the best result in group settings, and underline indicates the second-best result.

loss functions from different perspectives. Therefore, the overall image-level enhancement KD loss is formulated as:

$$\mathcal{L}_{\text{IeKD}}=\begin{cases}\mathcal{L}_{\text{SKD}}=\mathcal{L}_1(P_j,P_j'), & Q_j=0\\ \mathcal{L}_{\text{CE}}(P_j,V_j), & Q_j=1\\ \mathcal{L}_{\text{CE}}(P_j,V_j)+\mathcal{L}_1(P_j,V_j)+\mathcal{L}_{\text{MSE}}(P_j,V_j), & Q_j=2\end{cases} \tag{5}$$

where P_j denotes the prediction of the j-th sample by the student model, and P_j' denotes the corresponding prediction after image augmentation. $\mathcal{L}_{\text{MSE}}$ represents the Mean Squared Error loss, and $\mathcal{L}_1$ denotes the L1 loss.

Pixel-Level Enhancement. We observe that most failures of MLLM stem from over-localization, meaning that the accuracy of the background outside the bounding box is largely unaffected by localization errors, resulting in pixels outside the box being highly reliable as background labels. Therefore, we construct a label S based on the bounding box, where only the background regions outside the box are labeled. S serves as additional supervision in the distillation process. In addition, the candidate masks also contain valuable pixel-level correctness priors. To exploit these, we compute an entropy map based on them:

$$\bar{V}=\frac{1}{3}(V^1+V^2+V^3), \tag{6}$$

$$E_i=-\bar{V}_i log(\bar{V}_i)-(1-\bar{V}_i)log(1-\bar{V}_i), \tag{7}$$

where i is the pixel index. The entropy map represents the uncertainty of each pixel. Following the principle that more stable pixels are more reliable, we invert the entropy map to generate a weight map $M_i=1-E_i$. By combining the pixel-prior S and M, we obtain the final loss function:

$$\mathcal{L}_{\text{GKD}}=\sum_i \mathcal{L}_{\text{IeKD}}(P_i,V_i)*\mathrm{M}_i+\sum_{i\in\tilde{S}}\mathcal{L}_{\text{IeKD}}(P_i,S_i), \tag{8}$$

where $\tilde{S}$ represents the labeled area of S, $*$ denotes an element-wise multiplication.

4. Experiments

4.1. Experimental Setup

Dataset. We evaluate our UCOD-MKD on three benchmark datasets, CAMO, COD10K, and NC4K. Following previous studies [11], we use a training set consisting of 1,000 images from CAMO and 3,040 images from COD10K, with the remaining images reserved for testing.

Evaluation Metrics. We adopt four evaluation metrics: mean absolute error (MAE), S-measure (S_m) [9], E-measure (E_m) [10], weighted F-measure (F_β^w) [31].

Implementation Details. We conduct experiments on an NVIDIA RTX A6000 GPU. Qwen2.5-VL-3b [2] and ViT-H are adopted as MLLM and SAM. For the backbone network, we employ PVT V2 [43]. During training, we use

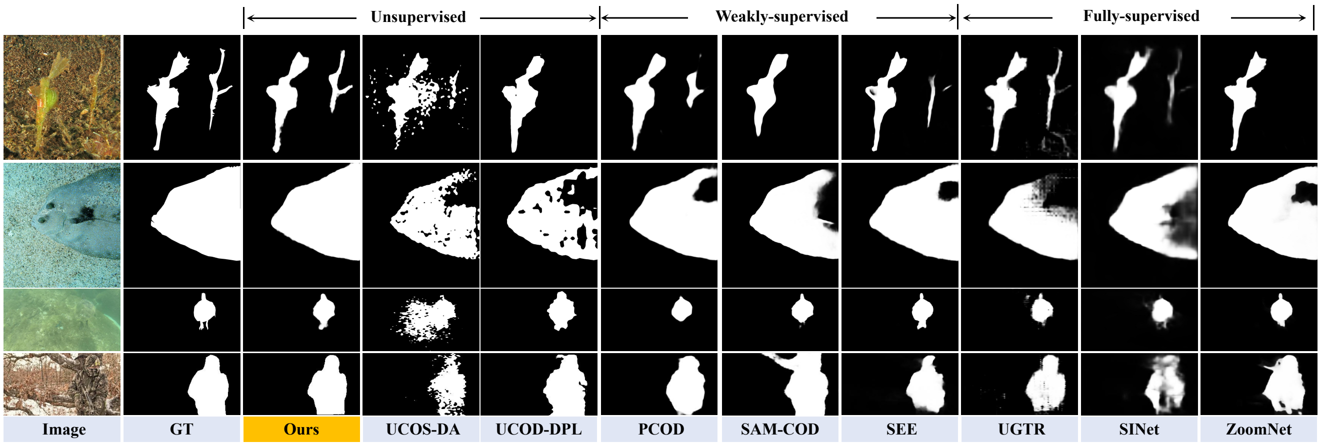

Figure 6. Visual comparison with state-of-the-art fully-supervised, weakly-supervised and unsupervised models.

Methods	Sup.	Params	MACs	FPS	COD10K			NC4K		
					MAE↓	E_m↑	F^w_β↑	MAE↓	E_m↑	F^w_β↑
PCOD [5]	P	62.6	52.6	31.2	0.042	0.859	0.650	0.051	0.889	0.748
SAM-COD [6]	S	62.6	52.6	31.2	**0.029**	0.904	0.728	**0.039**	0.912	0.795
UCOD-MKD (Ours)	U	60.3	50.7	34.8	0.031	**0.908**	**0.740**	0.040	**0.918**	**0.803**

Table 2. Comparison of parameters, MACs and FPS.

MLLM	CA-CoT	SAM	GME	GKD	CAMO MAE↓	CAMO E_m↑	COD10K MAE↓	COD10K E_m↑	NC4K MAE↓	NC4K E_m↑
✓		✓			0.205	0.711	0.232	0.685	0.176	0.761
✓	✓	✓			0.145	0.777	0.085	0.807	0.090	0.847
✓	✓	✓	✓		0.081	0.862	0.041	0.868	0.049	0.883
✓	✓	✓	✓	✓	**0.071**	**0.875**	**0.031**	**0.908**	**0.040**	**0.918**

Table 3. Ablation study of UCOD-MKD.

an SGD optimizer with a momentum of 0.9, a weight decay of $5e$-4, and triangle learning rate schedule with maximum learning rate $1e$-3. The batch size is 8, and the training epoch is 60. The entire training process takes around 7 hours. During both training and inference, input images are resized to 512×512.

4.2. Compare with State-of-the-arts

Quantitative analysis. As shown in Tab. 1, we compare our method with state-of-the-art methods under both unsupervised and zero-shot settings: 1) In the unsupervised setting, our method eliminates the dependency on the unsupervised backbone network DINO [4]. Compared to UCOD-DPL [45], our method achieves average gains of 42.6% (MAE), 14.0% (S_m), 9.2% (E_m), and 22.5% (F^w_β); 2) In the zero-shot setting, our method achieves state-of-the-art performance with fewer parameters (∼3B) and requires only two foundation models.

Qualitative analysis. Our method generates prediction maps with clearer and more complete object regions, along with sharper contours, as shown in Fig. 6. This advantage holds across various challenging scenarios, including: 1) tiny objects (row 3), 2) high intrinsic similarities (row 2), (3) poorly defined boundaries (row 4), and (4) complex backgrounds (row 1).

STEP1	STEP2	STEP3	STEP4	CAMO MAE↓	CAMO E_m↑	COD10K MAE↓	COD10K E_m↑	NC4K MAE↓	NC4K E_m↑
				0.205	0.711	0.232	0.685	0.176	0.761
✓				0.184	0.729	0.197	0.716	0.149	0.786
✓	✓			0.175	0.741	0.173	0.731	0.137	0.802
✓	✓	✓		0.161	0.758	0.133	0.766	0.118	0.823
✓	✓	✓	✓	**0.145**	**0.777**	**0.085**	**0.807**	**0.090**	**0.847**

Table 4. Ablation study of the CA-CoT module.

Methods	Extra Cost	CAMO MAE↓	CAMO E_m↑	COD10K MAE↓	COD10K E_m↑	NC4K MAE↓	NC4K E_m↑
SAM-COD [6]	1× Training	0.117	0.816	0.057	0.840	0.064	0.873
SEE [19]	11× Inference	0.102	0.822	0.059	0.831	0.067	0.864
Baseline	0	0.145	0.777	0.085	0.807	0.090	0.847
Baseline+SIM	∼ 0	0.120	0.812	0.066	0.836	0.070	0.870
Baseline+SIM+FGC	∼ 0	0.097	0.833	0.054	0.849	0.057	0.876
Baseline+SIM+ETC	∼ 0	0.092	0.840	0.052	0.852	0.054	0.874
Baseline+SIM+FGC+ETC	∼ 0	**0.081**	**0.862**	**0.041**	**0.868**	**0.049**	**0.883**

Table 5. Ablation study of the GME module.

Computational cost analysis. As shown in Tab. 1, although our model leverages foundation models (FMs), it requires fewer and smaller FMs compared to other zero-shot methods. Moreover, inference is performed just once, whereas ProMaC [24] and GenSAM [23] require six and twelve iterations, respectively. In the unsupervised training setting, the inference of FMs is decoupled from the training of the backbone. Since the foundation model does not require training, its memory usage is about ∼11GB with an inference time of ∼2h, which is not much higher compared to the backbone training (memory usage ∼21GB and training time ∼7h). Moreover, we compare our method with the WSCOD methods across five key aspects: performance, supervision level, parameter count, MACs, and FPS. As shown in Tab. 2, our UCOD-MKD achieves competitive accuracy while requiring less computation and weaker supervision.

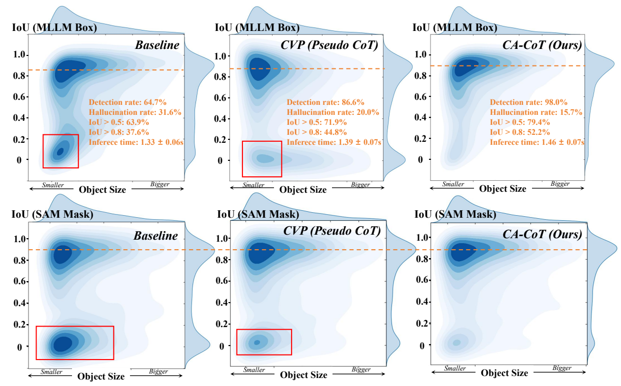

Figure 7. Joint density distribution of IoU versus camouflaged object size for MLLM-generated bounding boxes and SAM-generated masks. To ensure fair comparison, all experiments are conducted uniformly using Qwen2.5-VL$_{3B}$.

SSIM	IoU	CAMO		COD10K		NC4K	
		MAE↓	E_m↑	MAE↓	E_m↑	MAE↓	E_m↑
×	×	0.145	0.777	0.085	0.807	0.090	0.847
✓	×	0.129	0.793	0.073	0.822	0.076	0.861
×	✓	0.122	0.807	0.071	0.826	0.074	0.864
✓	✓	**0.120**	**0.812**	**0.066**	**0.836**	**0.070**	**0.870**

Table 6. Ablation study of SIM.

SIM	Num	TN (%)	FN (%)	ETC	Num	TN (%)	FN (%)	FGC	Num	TN (%)	FN (%)
	823	82.7	17.3		541	90.0	10.0		182	95.1	4.9

Table 7. Statistics of GME low-quality mask identification. TN denotes true negative, FN denotes false positive. Negative samples are determined by computing IoU between the mask and the ground truth, with an IoU threshold of 0.7 (derived from Fig. 7)

4.3. Ablation Study

Effect of CA-CoT. As shown in Tab. 3, CA-CoT contributes significantly to performance improvement. We compare the proposed CA-CoT with both the baseline and CVP [40]. As shown in Tab. 4, we perform a detailed ablation of CA-CoT, and each step is effective. As shown in Fig. 7, CA-CoT achieves superior localization quality, exhibiting higher IoU, improved detection rate (camouflaged object detected), and reduced hallucination rate, while obtaining the box as a prompt significantly enhances the mask quality of subsequent SAM outputs. Notably, although CA-CoT adopts step-by-step Chain-of-Thought (CoT) reasoning, it achieves nearly a 10% performance improvement with only about a 5% increase in inference cost, demonstrating the effectiveness of its reasoning design.

Effect of GME. We conduct ablation experiments for GME. As shown in Tab. 5, each component of GME contributes significantly to performance improvement. As shown in Tab. 6, we ablate the components of SIM, verifying the effectiveness of its design. In addition, we conduct experiments on the discrimination capability of GME. As shown in Tab. 7, GME can accurately identify erroneous masks. We further evaluate GME against existing SAM mask processing methods. As shown in Tab. 5, GME delivers superior performance with negligible additional overhead. The sensitivity analysis of hyperparameters in GME is detailed in the supplementary material.

Baseline	S	M	Q_j=0	Q_j=2	CAMO		COD10K		NC4K	
					MAE↓	E_m↑	MAE↓	E_m↑	MAE↓	E_m↑
✓					0.081	0.862	0.041	0.868	0.049	0.883
Pixel-Level Enhancement										
✓	✓				0.077	0.870	0.039	0.866	0.048	0.897
✓		✓			0.078	0.867	0.041	0.873	0.048	0.893
✓	✓	✓			0.074	0.874	0.038	0.877	0.047	0.901
Image-Level Enhancement										
✓			✓		0.078	0.869	0.038	0.877	0.047	0.900
✓				✓	0.074	0.871	0.035	0.891	0.041	0.905
✓			✓	✓	0.072	0.872	0.032	0.904	0.040	0.913
✓	✓	✓	✓	✓	**0.071**	**0.875**	**0.031**	**0.908**	**0.040**	**0.918**

Table 8. Ablation study of the GKD module.

L1	MSE	CAMO		COD10K		NC4K	
		MAE↓	E_m↑	MAE↓	E_m↑	MAE↓	E_m↑
×	×	0.081	0.862	0.041	0.868	0.049	0.883
✓	×	0.078	0.866	0.038	0.879	0.046	0.891
×	✓	0.076	0.869	0.036	0.884	0.044	0.898
✓	✓	**0.074**	**0.871**	**0.035**	**0.891**	**0.041**	**0.905**

Table 9. Ablation study of image-level enhancement at $Q = 2$.

Effect of GKD. We conduct an ablation study on GKD. As shown in Tab. 8, both pixel-level enhancement and image-level enhancement contribute significantly to performance improvement. As shown in Tab. 9, we test image-level enhancement at $Q = 2$, showing both L1 and MSE losses are effective. Notably, the image-level enhancement leads to a substantial boost in performance, demonstrating the importance of the quality grade of pseudo-labels, which enables the model to fully exploit and utilize pseudo-labels.

5. Conclusion

Existing UCOD methods face two main challenges: 1) Limited ability to obtain effective supervision, resulting in reliance on self-supervised backbones like DINO; 2) Inefficient use of pseudo-labels, leading to a large performance gap from supervised models even with strong backbones like DINO. To address these limitations, we propose UCOD-MKD with two key innovations: 1) To generate effective supervision, we construct a robust teacher model by integrating foundation models (MLLM and SAM) and design CA-CoT and GME to adapt these models specifically to the COD task. 2) To maximize supervision utilization, we propose GKD, which synergizes pixel-level and image-level distillation, significantly improving the performance of distillation. Experimental results show that our method significantly outperforms existing unsupervised approaches and even surpasses many weakly supervised methods.

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